# A New Method of Calculating the Number of Prime Numbers less than a Given Quantity

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## Abstract

This is a presentation of new procedures for calculating the number of prime numbers up to x by summation of prime powers. The idea is that the natural numbers count themselves, the prime numbers in some way do as well. Besides, the procedures are compared with existing formulas. In addition to the 1st edition the prime twins and k-tuples are now included.


## Introduction

The object of this paper is to present a new method to determine the "Anzahl der Primzahlen unter einer gegebenen Größe" (Riemann), which means the number of primes less than a given x. This is not about competition with existing procedures, but only to increase knowledge.

Let $\pi(x)$ be the familiar name of the (exact) number of prime numbers less than or equal to x. Let $\log x$ be the natural logarithm of x.

For the calculation of $\pi(x)$ there are approximate formulas, of which

| | |
|---|---|
| $\pi(x) \sim x / \log x$ | prime number theorem, PNT, Gauss |
| $\pi(x) \sim x / (\log x - 1)$ | Legendre etc. |
| $\pi(x) \sim \mathrm{Li}(x)$ | logarithmus integralis, Gauss |
| $\pi(x) \sim R(x)$ | Riemann |

are the best known ones. To explain the formulas see chapter 6. Since

$$\log x \approx x \cdot \left(\sqrt[x]{x} - 1\right),$$

the PNT can be written in the form

$$\pi(x) \sim \frac{1}{\sqrt[x]{x} - 1}$$

and the logarithmus integralis

$$\mathrm{Li}(x) = \sum_{n=2}^{x} \frac{1}{\log n}$$

in the form

$$\mathrm{Li}(x) \approx \sum_{n=2}^{x} \frac{1}{n \cdot \left(\sqrt[n]{n} - 1\right)},$$

thus

$$\pi(x) \sim \sum_{n=2}^{x} \frac{1}{n \cdot \left(\sqrt[n]{n} - 1\right)} \cdot$$

There is full equality between the two forms, if the results are rounded to units. Usually Li(x) is written in integral form.

The character ~ means asymptotic equality, i.e. the quotient of both sides of the equation approaches 1 with increasing x. Yet the (absolute) *difference* of the sides of the equation goes to infinity, which is usually kept secret and gives the approximation formulas only fundamental importance. The formulas are suitable for the theoretical understanding and a rough estimate of dimensions, an accurate evaluation of the number of primes cannot be expected. If authors speak of "increasing accuracy with increasing x", this means always *relative* accuracy. Thus the formulas offer little (absolute) accuracy at low computing time.

From formulas in the strict sense of the term expressions have to be distinguished which contain e.g. continuous summation or multiplication ($\Sigma$, $\Pi$) and which should rather be called "algorithms" or "calculating prescriptions". Examples are the logarithmus integralis and the Riemann formula. The Hardy algorithm (1979) counts the prime numbers even exactly. Upon closer inspection these algorithms are more or less a counting. The "New Method" is a mixture.

## 1. The original form

The "ideal" algorithm to determine the number of prime numbers is to count them. Here we have highest precision coupled with greatest time expenditure. If it may be a little more inaccurate, then it will work faster. At this point a contribution by "Sabine" in the forum of the German "Primzahlenseite" was useful, who posted the surprising statement (not literally): The number of prime numbers less or equal x is asymptotically equal the sum of the prime numbers up to the square root of x. In other words:

*Statement.* The sum of the prime numbers up to x equals asymptotically the number of prime numbers up to $x^2$:

(1.1)
$$\sum_{2}^{x} p \sim \pi\left(x^2\right)$$

Here p is less or equal x. The formula gets more exact if x is prime:

(1.2) $$\sum_{i=1}^{n} p_i \sim \pi\left(p_n^2\right)$$

Some examples show the truth of the statement:

| x | $\sum_{2}^{x} p$ | $\pi(x^2)$ | $\sum/\pi$ |
|---|---|---|---|
| $10^1$ | 17 | 25 | 0.68 |
| $10^2$ | 1060 | 1229 | 0.8624898291... |
| $10^3$ | 76127 | 78498 | 0.9697954088... |
| $10^4$ | 5736396 | 5761455 | 0.9956505778… |
| $10^5$ | 454396537 | 455052511 | 0.9985584653... |
| $10^6$ | 37550402023 | 37607912018 | 0.9984708008... |
| $10^7$ | 3203324994356 | 3204941750802 | 0.9994955426… |
| $10^8$ | 279209790387276 | 279238341033925 | 0.9998977551... |

Thus there is asymptotic equality between the approximate and the actual number of prime numbers.

*Proof.* Reflecting the assumption I noticed its similarity to

(1.3) $$\sum_{k=1}^{n} (2k-1) = n^2$$

I.e.: The sum of the odd numbers up to n equals n². You can also say: The sum of the odd numbers up to n equals the number of the natural numbers up to n². The natural numbers count themselves. Do the prime numbers count themselves in a certain way, too? At any rate there is also a relationship between sum and quantity and the increase of the degree by 1 (n to n²). The difference compared with the formulas (1.1) and (1.2) consists in the difference between odd numbers and prime numbers only.

As a proof we start from the prime number theorem introduced above: π(x) ~ x/log x. It has been proven several times (first in 1896 by Hadamard and de la Vallée-Poussin).

Furthermore one can assume the (arithmetically) average size pø of the primes up to x to be

(1.4) $$p\phi \sim \frac{x}{2}$$

The exact value of pø is somewhat smaller than x/2, but the difference is not very important, since pø approximates x/2 asymptotically with increasing x.

Now the sum of the prime numbers up to x simply results from multiplication of the number of primes by the average size:

(1.5) $$\sum_{2}^{x} p \sim \frac{x}{\log x} \cdot \frac{x}{2} = \frac{x^2}{2 \cdot \log x} = \frac{x^2}{\log x^2} \sim \pi\left(x^2\right)$$

According to PNT the outcome is the number of primes up to $x^2$, which had to be proved.

Another proof starts from the summation formula of the natural numbers

$$\sum_{k=1}^{x} k = \frac{x^2}{2} + \frac{x}{2} \cdot$$

We „shorten“ it logarithmically according to PNT, i.e. we elect the π(x) prime numbers from the x natural numbers. Its quantity is about x/log x. The right side is then shortened by the same ratio, too. Previously, we omit the summand x/2, since it is dispensable for the rough order. We receive once more:

(1.6) $$\sum_{2}^{x} p \sim \frac{x^2}{2} \cdot \frac{1}{\log x} = \frac{x^2}{\log x^2} \sim \pi(x^2).$$

## 2. Enlargement and generalization: Sand's summation theorem

As already indicated, sum and number of a polynomial of natural numbers are related to each other dimensionally. If the polynomial has degree n, the interval for counting the numbers will have degree n+1. This finds its clearest expression in the integration formula

(2.1) $$\int x^n \, dx = \frac{x^{n+1}}{n+1} \quad \text{for} \quad n \neq -1$$

The sum formulas themselves show this relationship, too:

(2.2)
$$\sum_{k=1}^{n} k = \frac{n^2}{2} + \frac{n}{2}$$

$$\sum_{k=1}^{n} k^2 = \frac{n^3}{3} + \frac{n^2}{2} + \frac{n}{6}$$

$$\sum_{k=1}^{n} k^3 = \frac{n^4}{4} + \frac{n^3}{2} + \frac{n^2}{4}$$

$$\sum_{k=1}^{n} k^4 = \frac{n^5}{5} + \frac{n^4}{2} + \frac{n^3}{3} - \frac{n}{30}$$

$$\sum_{k=1}^{n} k^m = \frac{n^{m+1}}{m+1} + \frac{n^m}{2} + O\left(n^{m-1}\right)$$

The last, general formula contains a residual member which consists of m-1 summands with binomial coefficients and Bernoulli numbers which we will prefer to avoid. As I said, we always understand the right side of the equations as the quantity of natural numbers.

It seems reasonable to form the correspondences for prime numbers:

(2.3)
$$\sum p = O\left(\pi(p^2)\right)$$

$$\sum p^2 = O\left(\pi(p^3)\right)$$

$$\sum p^3 = O\left(\pi(p^4)\right)$$

and so on, generally:

$$\sum p^n = O\left(\pi(p^{n+1})\right)$$

It is only one more step towards the generalization of statement (1.1) (“Sand’s summation theorem”):

The sum of the nth powers of primes less or equal x is asymptotically equal to the number of primes less or equal $x^{n+1}$:

(2.4) $$\sum_{p=2}^{x} p^n \sim \pi\left(x^{n+1}\right)$$

Or: The number of prime numbers up to x is asymptotically equal to the sum of the nth powers of the prime numbers up to $x^{1/(n+1)}$:

(2.5) $$\pi(x) \sim \sum_{p=2}^{x^{1/(n+1)}} p^n$$

## 3. Examples and verifications

By the example $x=10^{16}$ the precision of formula (2.5) shall be examined in dependence of n:

| | | |
|---|---|---|
| $\sum_{p=2}^{10^{16}} 1 = \pi\left(10^{16}\right) =$ | 279.238.341.033.925 | exact value, „counting“ |
| $\sum_{p=2}^{10^{8}} p =$ | 279.209.790.387.276 | |
| $\sum_{p=2}^{10^{16/3}} p^2 =$ | 278.959.908.085.255 | |
| $\sum_{p=2}^{10^{4}} p^3 =$ | 279.351.181.974.090 | |
| $\sum_{p=2}^{10^{16/5}} p^4 =$ | 282.984.841.298.409 | |
| $\sum_{p=2}^{10^{16/6}} p^5 =$ | 302.727.488.671.143 | |

Obviously (see first line) one can start already with n=0:

$$\sum_{p=2}^{x} p^0 = \sum_{p=2}^{x} 1 = \pi(x)$$

and gets the *exact* number of prime numbers, though at a maximum time expenditure ("counting"). Following the list downwards, the precision decreases, the required time decreasing as well. The list comes to an end at $10^{16/n+1} = 2$, i.e. at n = 52. Then $\Sigma p^{52} = 2^{52} = 4.503.599.627.370.496$. This is the fastest but most inaccurate "approximation". Between n=0 and n=52 the precision varies a lot.

For the practical use it may be considered with which n there is a reasonable balance between accuracy and time. This is partly a matter of taste. In our example I would choose n=3 or 4. The time expenditure decreases a lot between n=1 and n=3, after that only slightly. The difference at n=3 compared with the actual value (n=0) is 0.04 %.

Summary:

$$\pi(x) \sim \sum_{p=2}^{x^{1/(n+1)}} p^n$$

yields useful formulas at least for n=0,1,2,3,4 for determining the number of prime numbers below x (see also table in chapter 6). To what extent these observations can be generalized must still be investigated.

## 4. Proof of the summation theorem

*Theorem:*

$$\sum_{p=2}^{x} p^n \sim \pi\left(x^{n+1}\right) \quad n \in \mathbb{N}_0$$

*Proof:* We start from the summation formula (2.2) in its general form:

$$\sum_{k=1}^{n} k^m = \frac{n^{m+1}}{m+1} + \frac{n^m}{2} + O(n^{m-1}),$$

in which the 2nd and 3rd summand can be neglected without effect on the main order of the equation. n is changed to x, m to n:

(4.1) $$\sum_{k=1}^{x} k^n \sim \frac{x^{n+1}}{n+1}.$$

Again we shorten logarithmically according to PNT, i.e. we elect the $\pi(x) \sim x/\log x$ primes out of the x natural numbers (we multiply by the density):

(4.2) $$\sum_{p=2}^{x} p^n \sim \frac{x^{n+1}}{n+1} \cdot \frac{1}{\log x} = \frac{x^{n+1}}{\log(x^{n+1})} \sim \pi(x^{n+1}),$$

qed.

## 5. Further generalization

If we pass over from $n \in N_o$ to $r \in R_o^+$, we get to *theorem*

(5.1) $$\sum_{p \le x} p^r \sim \pi\left(x^{r+1}\right).$$

The proof goes by anology to n.

1st example: r = 1,5

| x | $\sum_{p<=x} p^r$ | $\pi\left(x^{r+1}\right)$ | $\sum/\pi$ |
|---|---|---|---|
| $10^1$ | 37.72517861 | 65 | 0.5803873632 |
| $10^2$ | 8143.985693 | 9592 | 0.8490393758 |
| $10^3$ | 1893442.417 | 1951958 | 0.9700221096 |
| $10^4$ | 453736185.7 | 455052511 | 0.9971073112 |

2nd example: r = log 2 = 0.6931471806...

| | | | |
|---|---|---|---|
| $10^1$ | 10.66242971 | 15 | 0.7108286473 |
| $10^2$ | 315.2843774 | 360 | 0.8757899372 |
| $10^3$ | 10971.55199 | 11307 | 0.9703327134 |
| $10^4$ | 405819.5106 | 407983 | 0.9946971089 |
| $10^5$ | 15827149.88 | 15854441 | 0.9982786451 |

Since we are dealing with *real* prime numbers and intervals, the possibility of generalizations ends here. David Broadhurst drew my attention to the fact that even negative r (r>-1) is possible and reasonable.

## 6. Comparison of certain procedures to determine the number of primes

The procedures compared:

$$\mathrm{Li}(x) = \sum_{2}^{x} \frac{1}{\log n}$$

$$R(x) = 1 + \sum_{n=1}^{\infty} \frac{\log^n(x)}{n \cdot n! \cdot \zeta(n+1)}, \text{ where } \quad \zeta(n) = \sum_{k=1}^{\infty} \frac{1}{k^n} \quad \text{is Riemann's zeta – function}$$

$$\pi(x) \sim \frac{x}{\log x} \qquad \text{prime number theorem}$$

$$\pi(x) \sim \frac{x}{\log x - 1} \quad \text{originates from Legendre, who still used the value } 1.08366 \text{ instead of } 1$$

$$S1(x) = \sum_{2}^{\sqrt{x}} p \qquad\qquad S2(x) = \sum_{2}^{\sqrt[3]{x}} p^2$$

The values in column 2. and 3. were rounded on units.
The values in column 1., 4., and 5. were taken from Sautoy p. 116.
The values in column 6. and 7. were calculated by MuPAD.

Under “comments“ the formula is given which comes nearest to S1 or S2. Thus S1 and S2 shall be put in the context of existing formulas. The fact that the Si tend to get more inaccurate with increasing i, has already been mentioned but can hardly be observed in this table. It is well known that in general x/log x provides the most inaccurate and R(x) the most accurate results.

**Comparison:**

| x | 1. π(x) | 2. x/log x | 3. x/(log x-1) | 4. Li(x) | 5. R(x) | 6. S1(x) | 7. S2(x) | Com. |
|---|---|---|---|---|---|---|---|---|
| $10^1$ | 4 | 4 | 8 | 6 | 5 | 5 | 4 | 5,2 |
| $10^2$ | 25 | 22 | 28 | 29 | 26 | 17 | 13 | 2,2 |
| $10^3$ | 168 | 145 | 169 | 177 | 168 | 160 | 87 | 5,2 |
| $10^4$ | 1.229 | 1.086 | 1.218 | 1.245 | 1.227 | 1.060 | 1.027 | 2,2 |
| $10^5$ | 9.592 | 8.686 | 9.512 | 9.629 | 9.587 | 9.206 | 10.466 | 3,4 |
| $10^6$ | 78.498 | 72.382 | 78.030 | 78.627 | 78.527 | 76.127 | 65.796 | 3,2 |
| $10^7$ | 664.579 | 620.421 | 661.459 | 664.917 | 664.667 | 642.869 | 609.586 | 3,2 |
| $10^8$ | 5.761.455 | 5.428.681 | 5.740.304 | 5.762.208 | 5.761.552 | 5.736.396 | 5.860.205 | 3,4 |
| $10^9$ | 50.847.534 | 48.254.942 | 50.701.542 | 50.849.234 | 50.847.455 | 50.481.978 | 49.345.379 | 3,2 |
| $10^{10}$ | 455.052.511 | 434.294.482 | 454.011.971 | 455.055.614 | 455.050.683 | 454.396.537 | 455.139.960 | 3,4 |
| $10^{11}$ | 4.118.054.813 | 3.948.131.654 | 4.110.416.301 | 4.118.066.400 | 4.118.052.495 | 4.108.749.732 | 4.048.935.205 | 3,3 |
| $10^{12}$ | 37.607.912.018 | 36.191.206.825 | 37.550.193.650 | 37.607.950.280 | 37.607.910.542 | 37.550.402.023 | 37.546.387.960 | 3,3 |
| $10^{13}$ | 346.065.536.839 | 334.072.678.387 | 345.618.860.221 | 346.065.645.810 | 346.065.531.066 | 345.842.863.160 | 343.650.297.652 | 3,3 |
| $10^{14}$ | 3.204.941.750.802 | 3.102.103.442.166 | 3.201.414.635.781 | 3.204.942.065.692 | 3.204.941.731.602 | 3.203.324.994.356 | 3.183.524.214.008 | 5,3 |
| $10^{15}$ | 29.844.570.422.669 | 28.952.965.460.217 | 29.816.233.849.001 | 29.844.571.475.288 | 29.844.570.495.887 | 29.841.977.261.093 | 29.822.760.083.883 | 5,3 |
| $10^{16}$ | 279.238.341.033.925 | 271.434.051.189.532 | 279.007.258.230.820 | 279.238.344.248.557 | 279.238.341.360.977 | 279.209.790.387.276 | 278.959.908.085.255 | 5,3 |

Both the S1 and S2 values seem to improve relatively with increasing x. In general S1 and S2 stand qualitatively between Legendre and Riemann. It is certainly an interesting task to extend the list by S3 and s4 values and up to higher x-areas.

## 7. Extending the method to prime twins

If we are to use a similar procedure for prime number *twins* as we used for "singles", we assume that their number is infinite, which for now must remain speculation, as a proof has yet to be found. The fact that there are formulas for the number of twins just as for single primes is not proof for infiniteness of their number. A prime number twin is defined by the span d=2 between adjacent prime numbers: d=q-p=2, p<q. This means that (2,3) is not a twin. The number of prime twins below x is represented by $\pi_2$ (x). Their simplest formula is:

(7.1) $$\pi_2(x) \sim \frac{x}{(\log x)^2}$$

This is easy to see looking at the prime number theorem if one simply postulates an additional prime number in the place of x. The overall probability for two prime numbers in x then is $1/(\log x)^2$. Strictly speaking, it is a bit more than that, because if x is a prime, either x+2 or x-2 can also be prime. This is taken into account by a 2 "prime twin constant" $c_2 = 0{,}66016\ldots$, so the formula is now, somewhat more precisely,:

(7.2) $$\pi_2(x) \sim 2 \cdot c_2 \cdot \frac{x}{(\log x)^2}$$

Corresponding to the simple primes, there are still other more precise formulas which to elaborate would go beyond the scope of this essay. Read, for instance, what Hardy and Littlewood have found on the number of k-tuples; it corresponds to the integral logarithm applying to simple primes. It should be noted at this point that the exactitude of Sand's summation theorem is somewhere between formula 7.2 and the formula by Hardy and Littlewood.

It still needs to be explained what it actually is we count when we count prime number twins. The twin is regarded as a unit, as a double, which means it is counted only once. As our intention is obviously to sum twins, we want to know the value of the individual twin. The smaller (p) or the larger (q) partner of the two or the arithmetic mean value (m) can be taken here. Which one we choose does not really matter for our scheme, but we have to decide, so we go for the impartial m. Thus follows that the sum of twins here is the sum of their mean values m=(p+q)/2.

Following these preparations, now the

*Conjecture.* The number of prime number twins below x is asymptotically equal to half the sum of the prime number twins below the square root of x:

(7.3) $$\pi_2(x) \sim \frac{1}{2} \cdot \sum_{4}^{\sqrt{x}} m$$

Again a few evidences, combined with a comparison with formula 7.1:

| x | $\pi_2(x)$ | $x/(\log x)^2$ | $\sum m/2$ ($m<\sqrt{x}$) | $\sum/\pi_2$ |
|---|---|---|---|---|
| $10^1$ | 2 | 2 | 0 | 0 |
| $10^2$ | 8 | 5 | 5 | 0.625 |
| $10^3$ | 35 | 21 | 35 | 1 |
| $10^4$ | 205 | 118 | 122 | 0.59512195… |
| $10^5$ | 1224 | 754 | 1322 | 1.08006535… |
| $10^6$ | 8169 | 5239 | 6059 | 0.74170645… |
| $10^7$ | 58980 | 38492 | 52631 | 0.89235334… |
| $10^8$ | 440312 | 294706 | 431603 | 0.98022084… |
| $10^9$ | 3424506 | 2328539 | 3310013 | 0.96656656… |
| $10^{10}$ | 27412679 | 18861170 | 27278642 | 0.99511040… |
| $10^{11}$ | 224376048 | 155877436 | 222430448 | 0.99132884… |
| $10^{12}$ | 1870585220 | 1309803451 | 1856091662 | 0.99225185… |
| $10^{13}$ | 15834664872 | 11160455444 | 15900559214 | 1.00416139… |
| $10^{14}$ | 135780321665 | 96230457659 | 136419649148 | 1.00470854… |
| $10^{15}$ | 1177209242304 | 838274208941 | 1176717615908 | 0.99958237… |
| $10^{16}$ | 10304195697298 | 7367644414516 | 10301443659233 | 0.99973292… |

I cannot yet offer a proof of the twin version of my theorem. The problem is the factor ½.

## 8. Extending the method to k-tuples

Let k be an arbitrary but firm natural number > 0. Then a k-tuple is a set of k consecutive prime numbers, where the difference between the greatest and the smallest is a minimum. The difference is even and produces more than 1 tuple. One assumes that then infinitely many tuples are produced. Thus e.g. (2,3) is not a twin and (3,5,7) not a triplet. The simple prime numbers are 1-tuples, the twins are 2-tuples. In this context we disregard the structure of the tuple, so e.g. (5,7,11) and (7,11,13) are simply triplets.

An expansion of the proceeding on k-tuples with k≧1 requires a factor c in

(8.1) $$\pi_k(x) \sim c \cdot \sum_{m<\sqrt{x}} m$$

where m is the arithmetic mean of the prime numbers concerned:

(8.2) $$m = \left( p_1 + p_2 + ... + p_k \right) / k$$

m might as well be any other value derived from the k-tuple.

c is assumedly

(8.3) $$c = \frac{1}{2^{k-1}} = 2^{1-k} \quad .$$

By analogy to the prime powers c is for certain verifiable for k=1 to 4; from k=5 upwards the results scatter considerably, understandable because of the increasing thinning of the tuple distribution.

We have still to investigate the summation of tuple powers starting from degree 2, as we did with the simple prime numbers. How useful is e.g.

(8.4) $$\pi_2(x) \sim \tfrac{1}{2} \cdot \sum_{m<\sqrt[3]{x}} m^2 \qquad ?$$

A verification test for $x=10^2$ to $10^{16}$ does not yield a clear result: the ratios $\sum/\pi_2$ are between 0.7 and 1.7, without a clear trend: a challenge for stronger computers and programs. The simple prime numbers already showed the ratios increasing with increasing prime power. I assume the k-tuples of degree 2 and more behave as well. Nevertheless we can theoretically state a generalized conjecture at least for the prime number twins:

(8.5) $$\pi_2(x) \sim \tfrac{1}{2} \cdot \sum_{m=4}^{x^{\frac{1}{n+1}}} m^n$$

Summarizing what has been said above we come to the overall conjecture:

(8.6) $$\pi_k(x) \sim \frac{1}{2^{k-1}} \cdot \sum_{m<x^{1/(r+1)}} m^r \qquad \text{or}$$

(8.7) $$\pi_k(x^{r+1}) \sim \frac{1}{2^{k-1}} \cdot \sum_{m<x} m^r$$

with real $r \geq 0$.

## 9. Closing remark

The reasonable summation of prime numbers resp. prime powers has certainly not come to an end. The procedure seems also to be applicable e.g. to “primeth primes” (prime numbers with prime index) or quite generally to “patterns”, which are repeating sequences of gaps between adjacent prime numbers. I ask interested people to participate in the subject with ideas and discoveries of their own. Furthermore I ask to tell me about maybe still existing faults in this script. If there are understanding questions, I will readily answer them. Above all I ask to point out other work in this field which has been done perhaps by other means and which have escaped me. I hope to have shown a little of the fascination which lies in the addition resp. summation of numbers actually made for multiplication, see Goldbach’s conjecture. Actually mathematics gets truly exciting not before being brushed the wrong way.

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